

Notice of Intent to Review Marketplace Solutions

SHOPPING NOTICE: DARPA-SN-26-103

MARKETPLACE: Defense Advanced Research Projects Agency (DARPA) Expedited Research Innovation System (ERIS)

REQUIRING ACTIVITY: DARPA, Defense Sciences Office (DSO)

NOTICE DATES: July 31, 2026 – September 30, 2026

TOPIC AREA: Developing advanced technologies for improved resilience, efficiency, and effectiveness of strategic systems. This topic area supports technologies for critical infrastructure and military command-and-control systems (networks and applications across strategic, command, operational, and tactical edges).

1.0 PURPOSE

Between the dates listed above, DARPA DSO will be actively “shopping” the ERIS Marketplace for game-changing, foundational mechanisms—rather than modular, integrated cooling packages—capable of fundamentally disrupting legacy thermodynamic limits across diverse DoD application spaces.

This is an active and immediate search for solutions. Submissions will be reviewed on a rolling basis in accordance with the procedures set forth in DARPA-PS-25-05.

2.0 PROBLEM & SCOPE

The advancement of critical DoD systems—including hypersonic vehicles, directed energy weapons, and high-performance tactical computing—has outpaced the development of viable thermal management solutions. Current state-of-the-art approaches are almost exclusively reliant on "brute force" methods, such as parasitic fluid pumping, forced convection, and sacrificial heat sinks. These legacy techniques are reaching a point of diminishing returns, where the cooling systems themselves impose unacceptable penalties in Size, Weight, Power, Cost, and Cooling (SWaP-C2), and present severe manufacturability and structural integration bottlenecks that restrict the performance and operational envelope of the systems they are intended to support.

DARPA is exploring the development of foundational technologies that enable a paradigm shift away from parasitic "bolt-on" cooling and toward novel, intrinsic thermal vectoring architectures.

- **Current Obstacles:** Existing thermal solutions are insufficient for future needs. Legacy methods that rely on active mechanical components (pumps, fans), parasitic bleed air, or dissimilar materials are fundamentally limited by reliability, efficiency, and physics constraints.
 - At the material level, high-conductivity materials (such as diamond, graphene, or ultra-high-temperature ceramics) are limited by high-temperature chemical instability, oxidation, or an inability to be fabricated into complex, load-bearing shapes without loss of structural properties.
 - At the structural level, cooling channels suffer from localized thermal-stress concentration, structural delamination, and mechanical failure driven by transient thermal shock and localized temperature gradients across load-bearing members.
 - At the thermodynamic level, heat pipes and phase-change loops are limited by capillary dry-out, vapor-choking, and fluid-materials incompatibility under acceleration, vibration, or transient heat-flux conditions.
- **Desired Research Focus:** We are seeking unconstrained concepts that treat thermal management as a design variable rather than a boundary condition. Key research and innovations are expected to focus on

developing novel methods and mechanisms to efficiently route, dissipate, or transduce extreme thermal loads. Proposals that are limited to incremental improvements of conventional cooling systems will be viewed unfavorably.

3.0 DESIRED CAPABILITIES AND EXPERTISE

While *all submissions will be evaluated against the standard criteria published in the ERIS Master Solicitation*, DSO will be specifically looking for solutions that demonstrate strength in the following areas. These are not new evaluation criteria, but rather highlight the specific attributes of interest for this opportunity.

- **Defining the Problem & Current State of the Art:**

Strong submissions will demonstrate a deep understanding of the problem space. They should clearly define the scope of the technical challenge they address and compare their proposed solution to the limitations of the current state of the art, particularly regarding the SWaP-C2 penalties of legacy conventional thermal management. Because SWaP-C2 constraints vary drastically across domains, proposers must explicitly define the target operational envelope of their chosen class of systems (e.g., hypersonic, space, directed energy, or edge computing) and clearly demonstrate how their proposed innovation fundamentally expands that specific system's performance boundaries.

- **Advancing the State of the Art (Technical Innovation):**

Performers should define their own aggressive performance metrics (e.g., effective thermal conductivity, specific heat transfer, mass-specific rejection, SWaP-C2) and provide clear technical insights into how their proposed innovation overcomes the limitations of existing solutions. Submissions must prioritize rigorous, physics-based evidence—such as mathematical derivations, first-principles thermodynamics calculations, or preliminary modeling data—over qualitative, marketing-style assertions. Proposals lacking concrete, quantifiable evidence to support their performance claims will be viewed unfavorably.

- **Team Capability (Expertise and Vision):**

We are seeking multidisciplinary teams with a strong vision and demonstrated expertise in discovering and validating new physical principles or generalizable approaches to heat transfer, rather than engineering point solutions for a single platform. Proposing teams should possess significant, combined expertise in relevant fields such as materials science, thermodynamics, multiphysics modeling, and/or advanced manufacturing. A proven track record of developing and validating novel physics-based concepts is highly desirable.

- **Impact (Defense and Commercial Use Case):**

Solutions should be broad enough to impact a range of potential systems or classes of systems as opposed to a single platform. Submissions should discuss how their innovation will enable future capabilities that are currently impossible due to thermal constraints across multiple operational domains for both defense and commercial industry applications.

4.0 HOW TO RESPOND

This notice is a call for solutions to be flagged for review.

For Existing Awardable Solutions:

If you have a solution already in the ERIS Marketplace that aligns with this need, send an email to ERIS@darpa.mil with:

- Subject Line: Shopping Notice Response: DARPA-SN-26-103 - [Your Company Name]
- Body: The Title and Video URL of your existing awardable solution.

For New Submissions:

If you have a new solution, please submit it to the "Strategic Systems" Topic Area, listed below, via the ERIS portal.

Developing advanced technologies for improved resilience, efficiency, and effectiveness of strategic systems. This topic area supports technologies for critical infrastructure and military command-and-control systems (networks and applications across strategic, command, operational, and tactical edges).

To ensure it is associated with this notice, please use the following format for your video title:

- Video Title Format: DARPA-SN-26-103: [Your Solution Title]
(Example: DARPA-SN-26-103: Thermal Management Solution Tool)

After submitting, please also send a notification email to ERIS@darpa.mil with:

- Subject Line: Shopping Notice Response: DARPA-SN-26-103 [Your Company Name]
- Body: The Title of your new submission and its Date of Submission.

Solutions submitted during this window that are deemed "awardable" through the standard ERIS assessment process will be included for consideration by DSO.

4.0 RESOURCES

- ERIS Master Solicitation: [SAM.gov Link](#)
- DARPAConnect offers free resources to potential performers to help them navigate DARPA, including “Understanding DARPA Award Vehicles and Solicitations”, “Making the Most of Proposers Days”, and “Tips for DARPA Proposal Success”. Join [DARPAConnect](#) at DARPAConnect.us to leverage on-demand learning and networking resources.
- To Submit a New Solution: [DARPAConnect ERIS Portal](#)